

Value Stream prediction — representation EDA

Question: for VS prediction, what text representation works best in each of the three places text is used — and can we drop summarization to save cost?

Setup (held constant across every run): - 100 tickets with ≥ 3 ground-truth VS (`--min-gt 3 --sample 100` , fixed seed) — the hard, discriminating cases; single-VS tickets wash out differences. - `--count-mode gt` — request exactly the GT count, so micro **F1 = P = R** (one clean number per run) and no count-generosity bias. - evidence mode, the Recall selection prompt, K=6 historic neighbours, judge on.

Terms & token counts (read this first)

Every IDMT ticket produces **two forms of text**. The whole study is about which form goes where.

term	what it actually is	size
raw	the ticket's description + extracted attachment text, consolidated and capped during ingest	$\leq 24,000$ tokens (<code>doc_char_budget = 96k</code> chars)
summary	the condense LLM's output from that raw — 6 fields: generatedSummary, businessProblem, businessCapability, keyTerms, stakeholders, systemsAndProducts	~ 460 tokens (all 6 concatenated)
description	only the ticket's Jira description field (no attachments)	a few hundred tokens
snippet	the first ~ 200 characters of a stored doc	~ 50 tokens
<code>raw@Nk</code>	the raw truncated to N-thousand tokens (e.g. <code>raw@7k</code> = first $\sim 7,000$ tokens of the raw)	N,000 tokens

These forms get used in **three independent places** per prediction:

place	what it does	what goes here (winner)
Retrieval query	the text we embed to search the index for the 6 nearest past tickets	the summary (~460 tok)
New-ticket prompt	the new ticket's text the LLM reads to pick the VS	the full raw (~24,000 tok)
Historic block	each of the 6 retrieved neighbours, shown as precedent in the prompt	each neighbour's summary (~460 tok × 6) + its VS labels

So a run labelled "**raw + summary**" in the tables below means: **new-ticket prompt = raw**, **historic block = summary** (the retrieval query stays the summary unless the row says `raw@7k` retrieval). The ladder varies one place at a time.

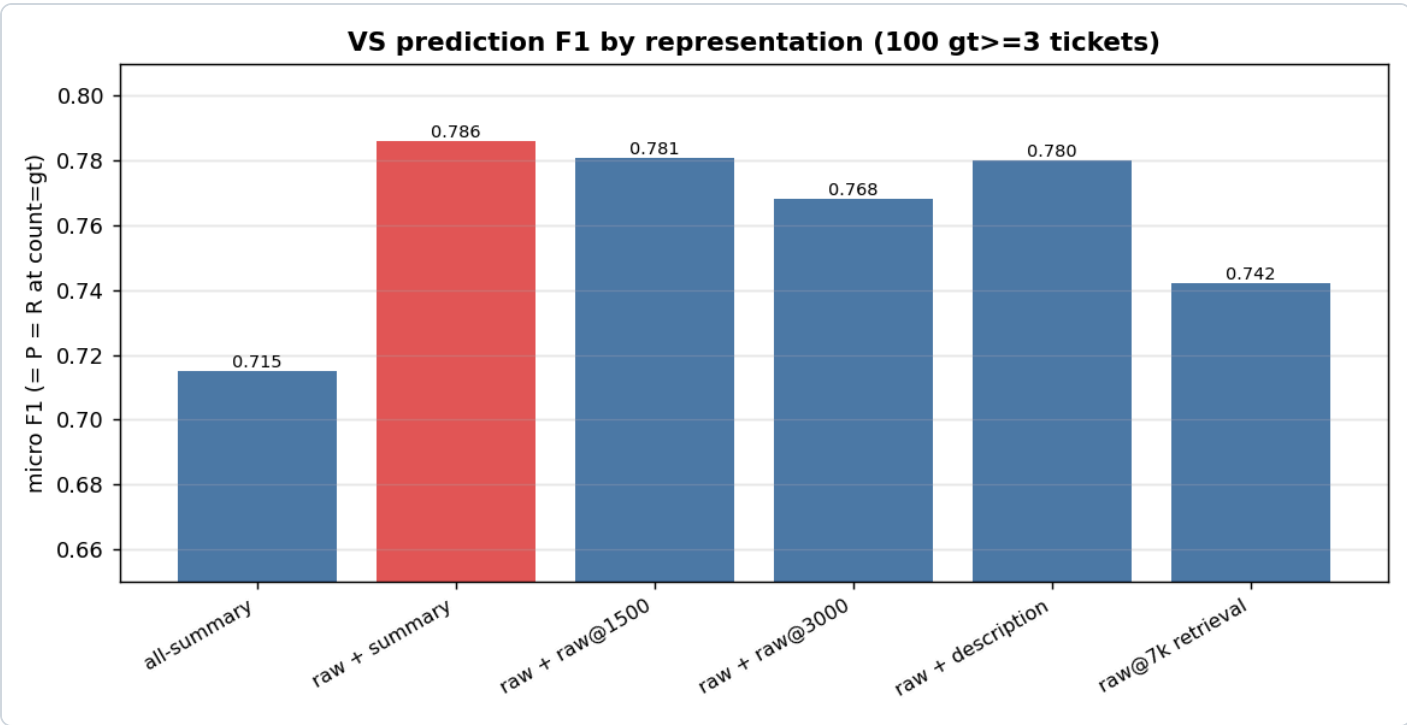
The new IDMT ticket is used in BOTH forms

This is the key thing to understand: for the **new** ticket we generate the summary **and** use the raw — they go to different places:

- its raw (~24k tok) is **summarized** → that **summary** (~460 tok) is what we **embed for the retrieval query** (to find the 6 nearest past tickets);
- the **same raw** (~24k tok, in full) is what we put in the **new-ticket prompt** (the text the LLM reads to pick the VS).

summary to *find*, raw to *decide*. The new ticket is therefore still summarized — that's why summarization can't be dropped. The 6 historic neighbours contribute **only their summaries** (their raw is never used in the winner).

Results



Columns = the form used in each of the three places (see the terms table above).

The **"new ticket summarized?"** column is the one to watch: every row except the last **still summarizes the new ticket** (its summary is the retrieval query). So those rows are *not* summary-free configs — they only change how the new ticket's prompt and the historic block are rendered. The summary is dropped **only** in the last row (raw@7k retrieval).

run	new ticket summarized?	new-ticket prompt	historic block	retrieval query	F1 (= P = R)	avg latency
all-summary	yes	summary	summary	summary	0.715	4.2s
raw + summary	yes	raw	summary	summary	0.786	5.8s
raw + raw@1500	yes	raw	raw@1500	summary	0.781	5.6s
raw + raw@3000	yes	raw	raw@3000	summary	0.768	6.3s
raw + description	yes	raw	description	summary	0.780	4.4s

run	new ticket summarized?	new-ticket prompt	historic block	retrieval query	F1 (= P = R)	avg latency
raw + raw@7k	yes	raw	raw@7k	summary	0.780	9.4s
raw@7k retrieval	NO	raw@7k	raw@7k	raw@7k	0.742	6.6s

Don't be misled by the middle rows. `raw + raw@1500` (0.781) looks close to the winner, but it **still summarizes** — it just shows the historic neighbours as `raw@1500` instead of `summary`. It is *not* evidence that raw historic is good or that summarization is droppable; it only says the historic representation barely matters **when the summary is still doing the retrieval**. The real summary-free test is the **last row** (the only "NO"), and it **drops to 0.742** — the verdict on dropping summarization. (An earlier summary-free variant with **raw 7k tok** historic cost 13.9s avg / 132s max latency; cutting historic to 3k fixed the latency but not the quality.)

How to read it: the big step is `summary`→`raw` on the **new-ticket prompt** (0.715 → 0.786). After that, swapping the **historic block** representation barely moves F1 (0.768–0.786) — and the last bar (`raw@7k retrieval`) drops *below* the pack. So the prompt is the lever, the historic block is a wash, and raw retrieval is a regression.

Why F1 = P = R here (they're normally different): precision = correct ÷ **predicted count**, recall = correct ÷ **GT count**. We force the model to return **exactly the GT count** (`--count-mode gt`) and it complies (avg predicted 5.9 = avg GT 5.9), so predicted count = GT count → the two denominators are equal → **P = R**, and F1 (their mean) equals that too. Pin a *different* count (fixed 10, or GT+2) and they'd diverge — we used `count=gt` to get one honest number.

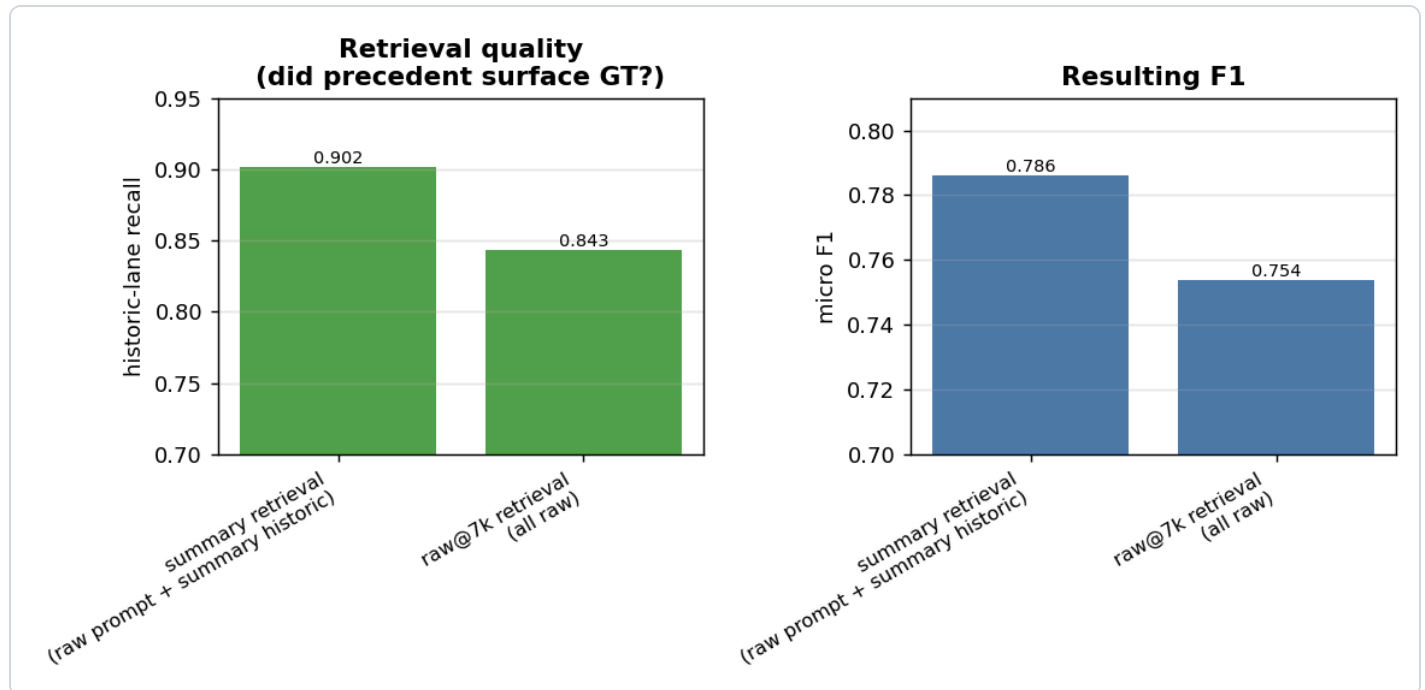
Finding 1 — the lever is the NEW-TICKET prompt

Feeding the new ticket's **raw text** instead of its summary is **+0.071 F1 (0.715 → 0.786)**. Retrieval is already perfect at surfacing candidates (every GT reaches the LLM), so all of this gain is the LLM *choosing* better when it sees the full ticket.

Finding 2 — the historic block representation is a wash (so use the cheapest)

Among the raw-prompt runs the historic block spans only 0.768–0.786: summary (0.786) $\approx$ description (0.780) $\approx$ raw@1500 (0.781) $\approx$ raw@7k (0.780), with raw@3000 slightly *worse* (0.768). More raw precedent does **not** help — it dilutes. **Summary historic is marginally best and the cheapest.**

Finding 3 — summary retrieval beats raw@7k retrieval (the cost question)

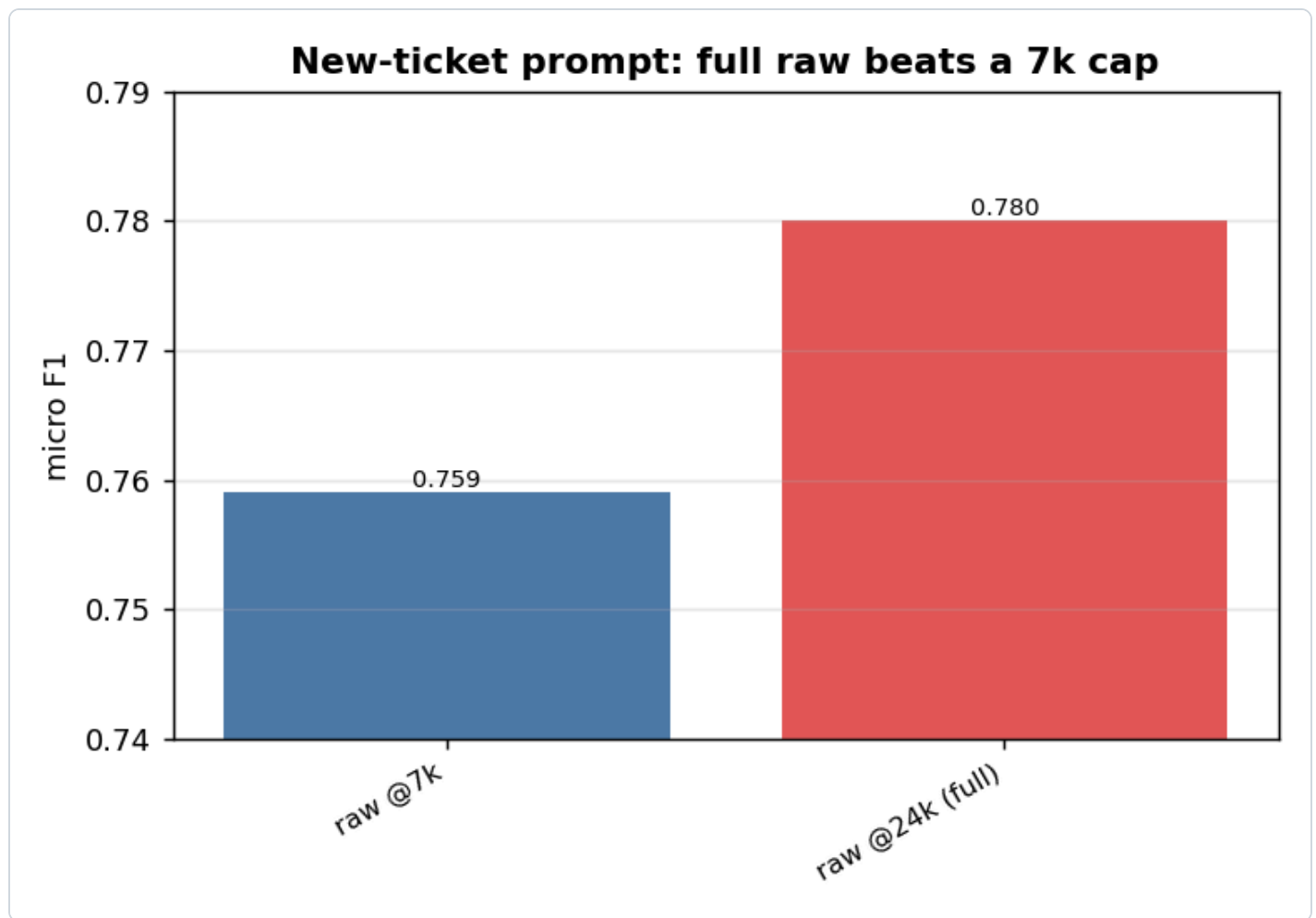


To test dropping summarization entirely, we re-embedded the index on **raw@7k** (no summary) and retrieved with raw. The decider is **historic-lane recall** — did the precedent search surface the GT (in evidence mode the 50-VS candidate pool makes review-pool recall structurally 1.0, so *that* isn't the signal — the historic lane is).

- summary retrieval: historic-lane R **0.902** $\rightarrow$ F1 **0.786**
- raw@7k retrieval: historic-lane R **0.843** $\rightarrow$ F1 **0.754**

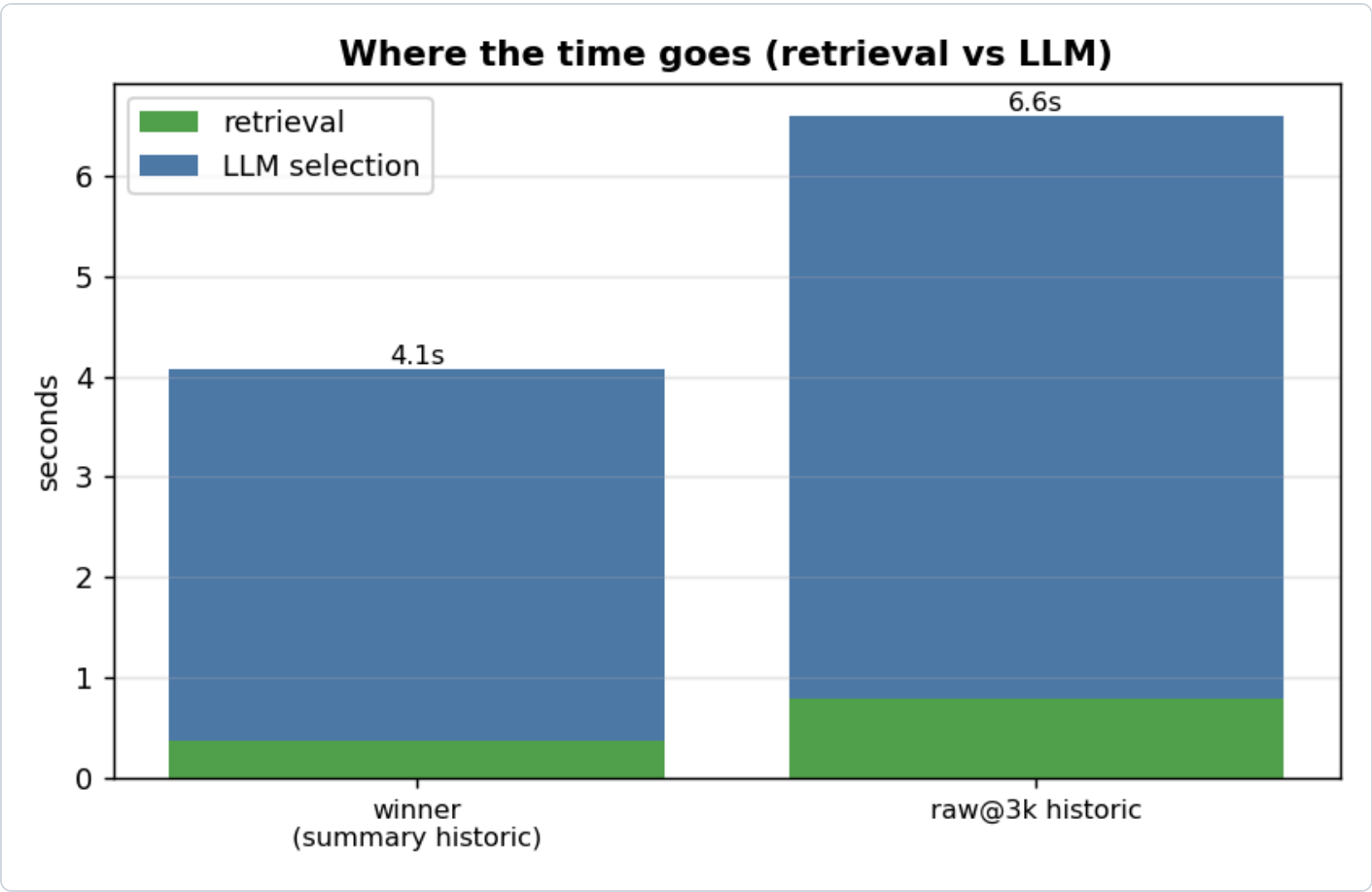
How to read it: the ~460-token summary embeds the *whole* ticket cleanly; raw@7k keeps only ~half of a big ticket, so it retrieves **worse neighbours**. Since precedent drives recall (GT *backed* by historic is picked ~0.82, *not-backed* only ~0.38), worse precedent $\rightarrow$ lower recall $\rightarrow$ ~3 F1 points lost. **Dropping the summary costs quality.**

Finding 4 — feed the new ticket's FULL raw; don't cap it



Holding everything else at the winner (summary retrieval + summary historic) and changing only the new ticket's raw cap: the **full ~24k raw** scores **0.780** vs **0.759** at a 7k cap — capping loses ~2 points of F1. The extra context genuinely helps the LLM decide, and latency was identical (~4s) either way, so **there's no reason to cap the new-ticket prompt**.

Where the time goes (latency split)



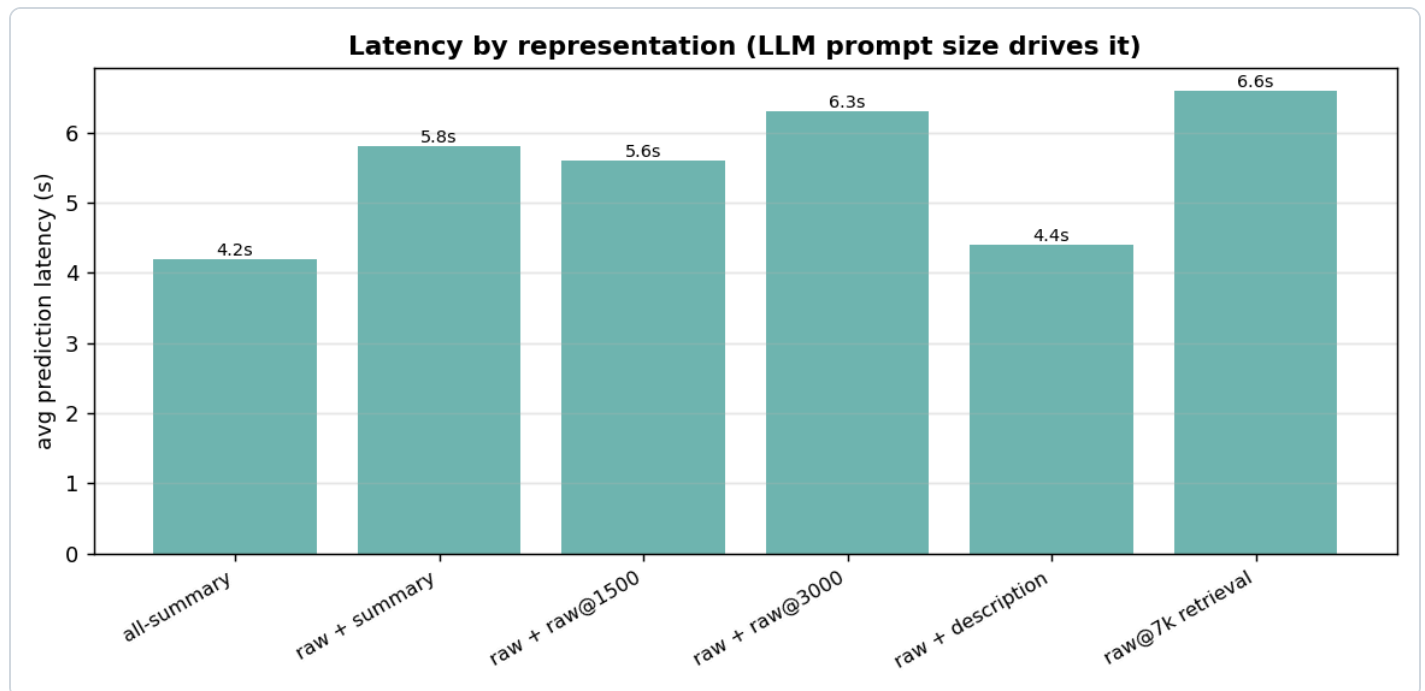
How to read it: splitting prediction latency into retrieval vs the LLM call shows **retrieval is sub-second** (0.38–0.79s) — never the bottleneck. The **LLM call is the whole cost**, and it tracks the **historic block size**: summary historic 3.7s vs raw@3k historic 5.8s. So the cheap lever for latency is the historic representation (keep it summary), not the new-ticket prompt.

Avg vs max latency (per-ticket, prediction only; the max is the slowest single ticket in the run):

config	new-ticket prompt	historic	avg	max
winner	raw ~24k tok	summary ~460 tok x6	3.7s	6.9s
capped winner	raw 7k tok	summary ~460 tok x6	3.7s	6.9s
fully-raw (confirmation)	raw 7k tok	raw 3k tok x6	5.8s	21.7s
raw@7k historic (earlier)	raw 7k tok	raw 7k tok x6	13.9s	132s

The winner's max is **6.9s** — tight, because the prompt is one ~24k-token ticket + six ~460-token summaries. The moment the historic block goes raw, the max blows out: **3k raw historic → 21.7s max, 7k raw historic → 132s max** (6 × 7k = 42k tokens of precedent → a ~55k-token prompt). Average hides this; the **max is what a user actually waits** on a bad-luck ticket — another reason to keep historic as summary.

Latency / cost



How to read it: latency tracks the LLM **prompt size**. summary historic $\approx 6 \times 460$ tokens; raw@7k historic $\approx 6 \times 7k = 42k$ tokens, which blows the prompt to $\sim 55k$ tokens and spikes prediction to 132s on big-neighbour tickets. This is a *second* cost axis, independent of the summary question: even if you keep summaries, **never ship raw@7k historic** — it's $\sim 4\text{--}5\times$ the runtime token cost for zero quality.

Count behaviour — does the LLM follow the requested count?

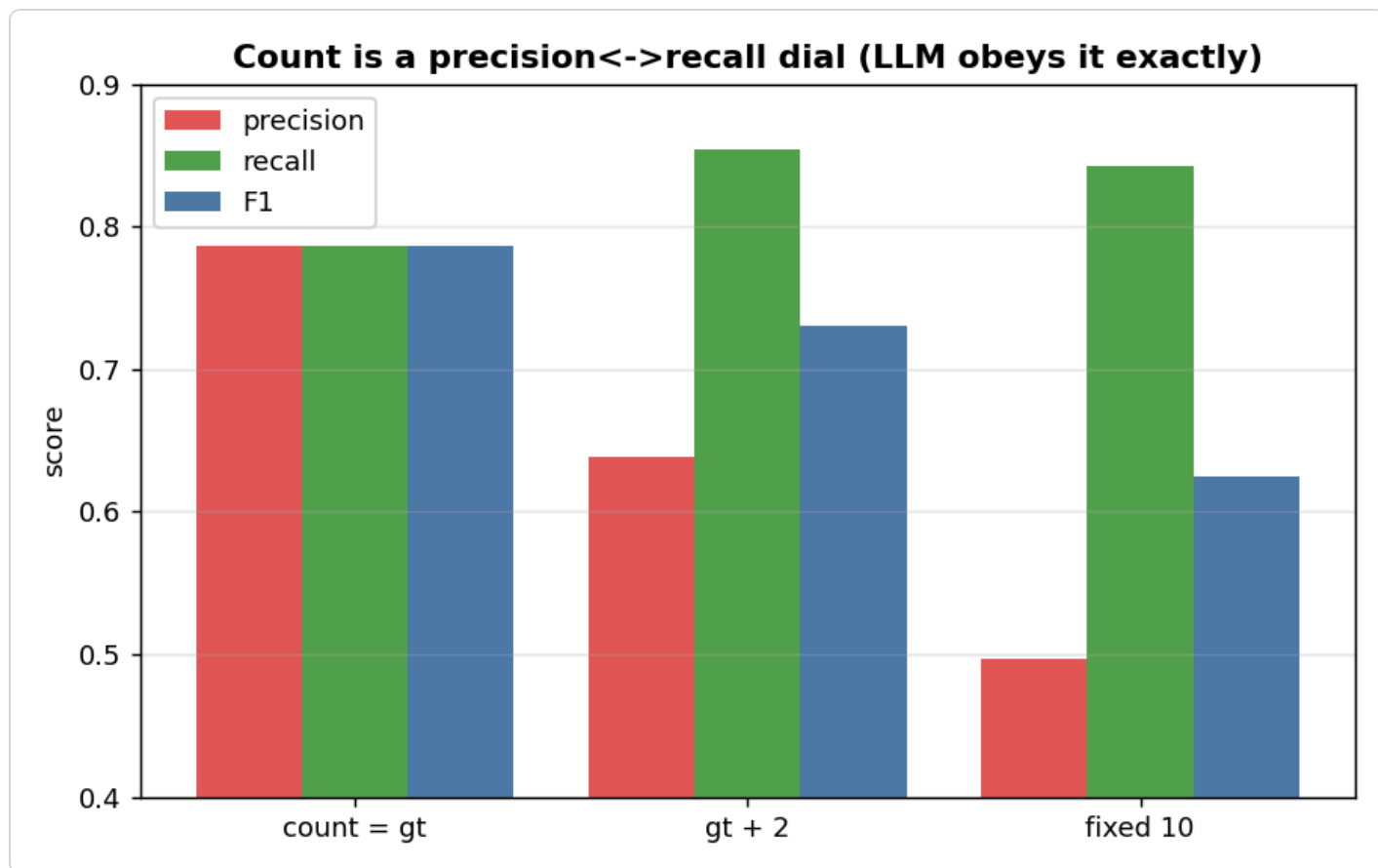
We pass a target count to the model. Two questions: (a) does it actually return that many on its own, and (b) when it skips a correct VS it *saw*, why? Run on the winner config, 100 $gt \geq 3$ tickets.

The LLM obeys the count exactly — to a fault

count requested	followed	padded by us	avg LLM picked
fixed 10	100%	0%	10.0
gt + 2	100%	0%	7.9

It returns **exactly** the requested count every time — we never had to pad. But it obeys *literally*: asked for 10 it returns 10 even when only ~ 6 are right, **filling the spare slots with weaker picks**.

So the count is a real, fully-obeyed **precision↔recall dial**:

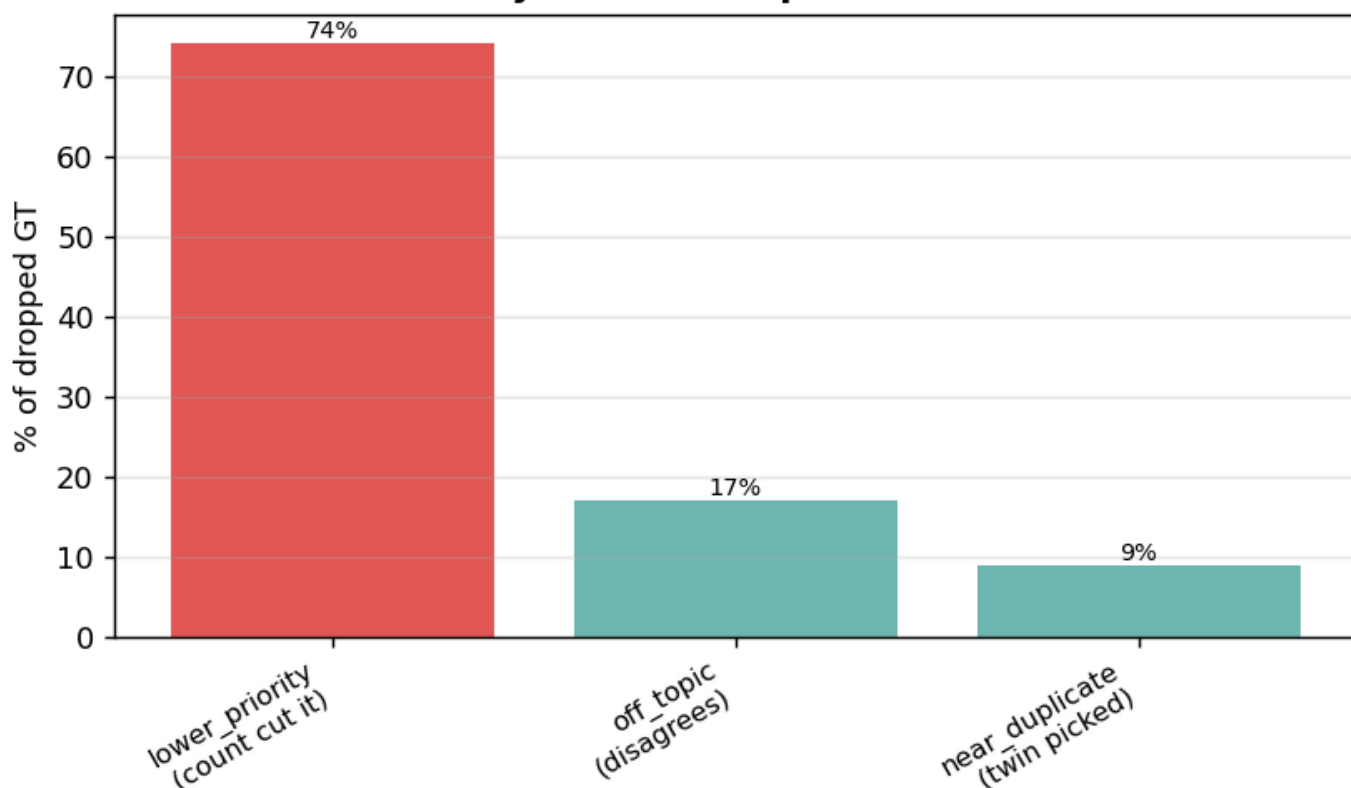


count mode	avg predicted	recall	precision	F1
count = gt	5.9	0.786	0.786	0.786
gt + 2	7.9	0.854	0.638	0.730
fixed 10	10.0	0.842	0.497	0.625

How to read it: more slots → recall climbs (0.786 → 0.854) but precision falls faster → F1 drops. Best F1 is at `count = gt`. Crucially, recall at `count = gt` is **genuinely capped** — giving +2 slots recovers ~7 points of recall, i.e. the model was dropping real GT just to fit the exact count.

Why it skips GT it saw — mostly the count, not quality

Why the LLM skips GT it saw



Every miss is `llm_dropped` (the GT reached the LLM — retrieval is perfect — and it didn't pick it). Re-prompting the model on each dropped GT buckets the reason:

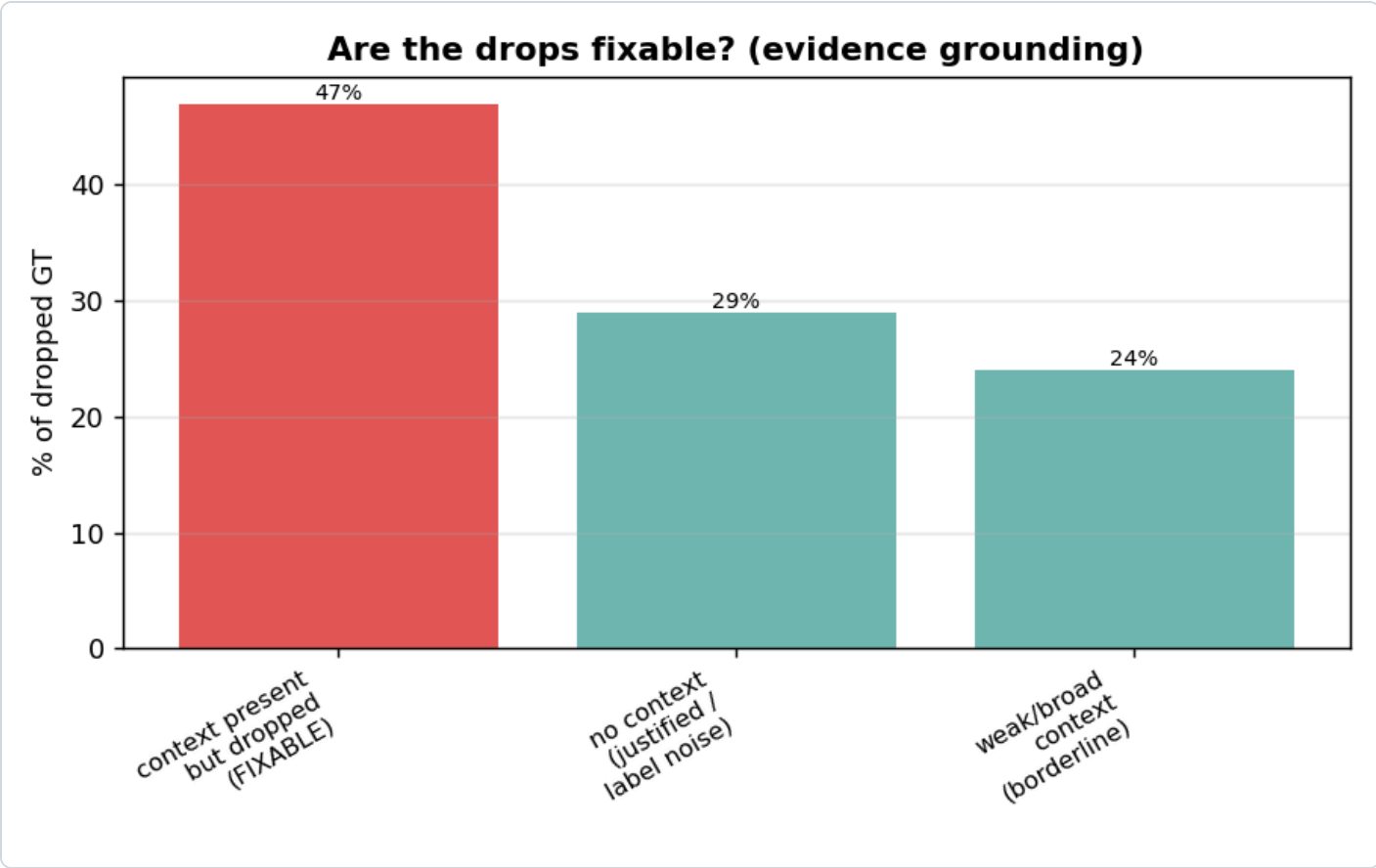
reason	share	meaning
lower_priority	~74%	relevant, but less central than the picks — the count squeezed it out
off_topic	~17%	the model genuinely disagrees with the GT label
near_duplicate_of_pick	~9%	a near-twin was picked instead

The two tests tie together: ~75% of misses are `lower_priority`, and Test 1 *proves* that's the count — give the model +2 slots and recall jumps $0.786 \rightarrow 0.854$, recovering exactly those VS. So **most "misses" aren't the model being wrong; they're the count constraint doing its job**. Only ~17% are genuine disagreement (and the judge says ~25% of *those* are actually supported — GT label noise). The genuinely fixable error surface is small (~17% off_topic + ~9% near-duplicate).

Implication: for the shortlist-then-trim product flow, pass **gt+2** — recall ~0.85, the model surfaces the relevant-but-lower-priority VS, and judge precision still holds (~0.71) so the extras are mostly fine. For best F1 / autonomous output, use **count = gt**.

Drop diagnosis — is the gap fixable, and can a prompt fix it?

We went deeper on the `count=gt` drops (131 of them, 100 $gt \geq 3$ tickets) with three probes: an evidence-grounding classifier, an independent 0-1 score of every candidate, and a comparative "why did the picks win" probe.



Grounding — how much is even fixable:

bucket	share	meaning
context_present_but_dropped	47%	the ticket supported it, the block showed it, dropped anyway — the fixable headroom
no_context_for_gt	29%	the ticket has no support — the BA tagged it from outside knowledge — not fixable from text (label noise)
weak_broad_context	24%	only indirect/downstream evidence — borderline

The model rates them relevant, then drops them. Scored *independently* (no count pressure), **81% of dropped GT scored $\geq$ a candidate the model actually kept**. So the model's relevance judgement is consistent; the single count-limited "pick N" call is where it goes wrong — it ranks

others higher and the count cutoff chops the rest. The comparative probe shows *what* it favours: the picks win as `picks_more_specific` (25%) / `picks_more_prominent` (8%), and the dropped GT lose as `dropped_too_broad` (39%) / `no_evidence_for_dropped` (27%). Only **2%** are misses the model itself calls genuine.

The root cause: the model is text-anchored; the BA is experience-anchored. The model rewards what the idea card spells out (specific, named, prominent) and under-weights the broad / downstream / precedent-implied streams — which is exactly the kind a Business Architect includes from experience. That is the core gap, and it's why precedent matters so much (GT *backed* by a similar past ticket is picked ~0.83 vs ~0.30 when unbacked): the historic block is the only thing that gives the model evidence for the broad/implied streams it would otherwise drop.

Prompt engineering — tried, and why it didn't move F1

We built `selection_evidence_recall_v2`, targeting the diagnosis directly: a **consistency** rule (don't drop a candidate as well-supported as your weakest pick — aimed at the 81% near-miss), a **broad/downstream reframe** (a broad stream is valid when the work flows through it; exclude only filler — aimed at the 39%+24%), and **near-twin care** (siblings can both apply). A/B on the same 100 tickets:

prompt	F1	P@6	R@6	judge F1
v1 winner	0.778	0.791	0.692	0.816
v2 (targeted)	0.776	0.803	0.705	0.829

Strict F1 did not move (0.776 vs 0.778, noise) — the rank/judge metrics ticked up a hair, the drop profile was unchanged. **Why:** the inconsistency is **structural, not promptable**. The model already *knows* the dropped GT is relevant (the 81% near-miss) — you cannot *talk* a single count-limited call into being consistent. Three prompt iterations would all land here. The lever is the **selection policy** (the count, or a two-stage score-then-select), not the wording.

Decision: stop here. Prompt wording is at its ceiling, and judge-F1 0.829 means ~ $\frac{1}{3}$ of the remaining "misses" are actually relevant (GT-label noise) — so true quality is already ~0.83. A two-stage *score-then-select* path was built (config `score_select`) as the one structural idea that could exploit the 81% near-miss, but we are **not pursuing it**: the realistic upside is small, the precision risk is real (generous scoring also lifts the magnet false-positives), and `count=gt` strict-0.78 / judge-0.83 is accepted as the corpus ceiling for selection-from-text. **VS selection is locked on the v1 winner.**

Verdict

Locked config: summary retrieval + FULL ~24k raw new-ticket prompt + summary historic — F1 ≈0.78–0.79.

- The summary you "can't eliminate" turns out to be the thing buying your best retrieval.
- Dropping summarization (raw@7k retrieval) is a **regression** (0.780 → 0.742, historic-lane R 0.902 → 0.840), not a marginal trade. The summary-free idea is dead.
- The new-ticket **raw** prompt is the real win (+0.07) — and feed it **in full**; a 7k cap costs ~2pts.
- Use the **cheapest** historic block (summary): same quality as raw, and it's what keeps the LLM call (the only real latency) at ~3.7s. Retrieval is sub-second regardless.

Final config (locked) — the per-ticket flow

1. New ticket raw, consolidated to ≤24k tokens
2. Summarize it (condense LLM) → summary fields
3. Embed the summary → search the index → retrieve the 6 nearest past tickets
4. Build the selection prompt:
 - the new ticket's FULL raw (~24k) ("raw to decide")
 - the 6 neighbours' summaries + their VS labels ("summaries as precedent")
 - the 50-VS catalogue
5. LLM picks the VS (count = requested; evidence mode, Recall prompt)

The two uses of "summary" are different field sets, by design: - **Retrieval query** embeds **all six** summary fields (generatedSummary + businessProblem + businessCapability + keyTerms + stakeholders + systemsAndProducts). - **Historic block** shows each neighbour's **generatedSummary only** + its VS labels.

So every new ticket contributes **both** its summary (to *find* precedent) and its full raw (to *decide*) — which is exactly why summarization stays in the pipeline.

Selection stays on the **v1 evidence-recall prompt at** `count=gt` (use `gt+2` only for the shortlist-then-trim flow). Prompt wording is at its ceiling; the drop gap is structural (text-anchored model vs experience-anchored BA) and ~1/3 label noise. **This EDA is closed — VS prediction is locked.**